

Assignment 3**Due April 9, 2026**

1. A two-level atom with a transition at angular frequency ω_0 and dipole moment $\mu_{12} = 2 \times 10^{-29} \text{ C m}$ is subjected to a sequence of two resonant pulses. The first pulse has an electric field given by $E(t) = E_1 \cos(\omega_0 t)$ and a duration of τ_1 , while the second has $E(t) = E_2 \cos(\omega_0 t + \phi)$ and a duration of τ_2 . E_1 and E_2 are both constant during the pulses, and the pulse length is much shorter than the dephasing time T_2 . Given that the system starts in the ground state, find the final state when:
 - (a) $E_1 = 4139 \text{ V m}^{-1}, \tau_1 = 1 \text{ ns}, E_2 = 6209 \text{ V m}^{-1}, \tau_2 = 2 \text{ ns}, \phi = 0$
 - (b) $E_1 = 827.8 \text{ V m}^{-1}, \tau_1 = 10 \text{ ns}, E_2 = 3311 \text{ V m}^{-1}, \tau_2 = 5 \text{ ns}, \phi = 90^\circ$
 - (c) $E_1 = 2.759 \times 10^4 \text{ V m}^{-1}, \tau_1 = 0.3 \text{ ns}, E_2 = 5.519 \times 10^4 \text{ V m}^{-1}, \tau_2 = 0.3 \text{ ns}, \phi = 45^\circ$

2. We derived following optical Bloch equations in terms of Bloch vector in class:

$$\frac{d}{dt} \begin{pmatrix} u \\ v \\ w \end{pmatrix} = - \begin{pmatrix} \Omega_0 \\ 0 \\ \delta \end{pmatrix} \times \begin{pmatrix} u \\ v \\ w \end{pmatrix} - \gamma \begin{pmatrix} u/2 \\ v/2 \\ w+1 \end{pmatrix}$$

- (a) Solve the above set of equations for $\Omega_0 = 0, \delta = 0$, and $\gamma > 0$.
 - (b) Assume the atom is initially in the excited state and sketch its path on the Bloch sphere for the above case. Calculate and draw the Bloch vector at $t = 0, t = \frac{\ln(2)}{\gamma}$, and $t = \infty$.
 - (c) Repeat the above exercise for initial state $u = 1, v = 0$, and $w = 0$.
 - (d) Assume a resonant field is present ($\Omega_0 > 0, \delta = 0$). What is the steady state value of the Bloch vector? Calculate and draw the steady state Bloch vector for $\Omega_0 = 0, \Omega_0 = \gamma, \Omega_0 = \infty$.
3. Consider a three-level atom interacting with a classical field of frequency ν . The transitions $|a\rangle \rightarrow |b\rangle$ and $|b\rangle \rightarrow |c\rangle$ are allowed whereas the transition $|a\rangle \rightarrow |c\rangle$ is forbidden. It is also assumed that $\omega_a - \omega_b = \omega_b - \omega_c = \nu$. Assuming the atom to be initially in level $|c\rangle$, find the probabilities for the atom to be in levels $|a\rangle$ and $|c\rangle$ after making the rotating wave approximation.